

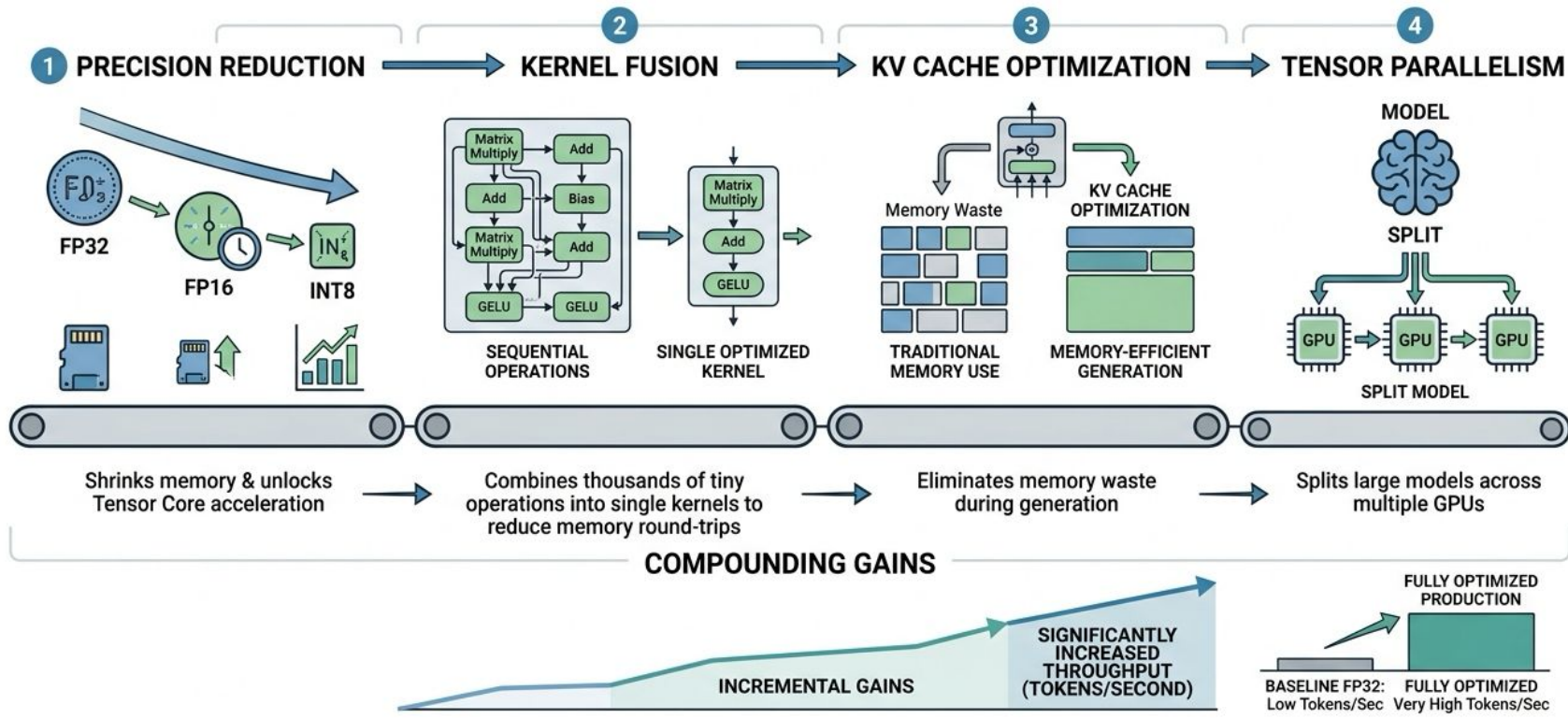


U.S. General Services
Administration

Chapter 4.6: TensorRT-LLM and NVIDIA Fleet Command

US General Service Administration
AI Community of Practice - March 2026

The Multi-Stage Optimization Pipeline

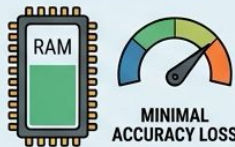


Quantization & the Calibration Catch

OPTIMIZING MODEL PRECISION

PRECISION OPTIONS

FP16 (HALF PRECISION)



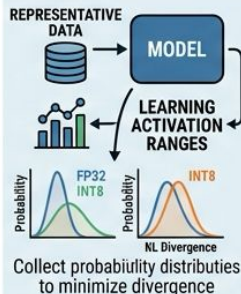
- Halves Memory Use
- Enables Tensor Cores
- High Accuracy

INT8 (8-BIT INTEGER)



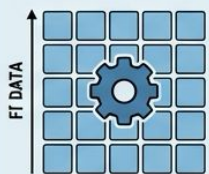
- 4x Speedup (Theoretical)
- Requires Calibration Steps

CALIBRATION METHOD



SCALING MODES

PER-TENSOR



FASTEST, COARSEST

PER-CHANNEL



BALANCE: SPEED & ACCURACY

PER-TOKEN



HIGHEST ACCURACY, ADAPTIVE

THE CRITICAL CALIBRATION DATASET

PRODUCTION INPUT DISTRIBUTION



✓ HIGH ACCURACY
INT8 MODEL

80% COMMON QUERIES

15% EDGE CASES

5% ADVERSARIAL INPUTS



GENERIC WEB TEXT



✗ POOR ACCURACY
(15-20% DEGRADATION)

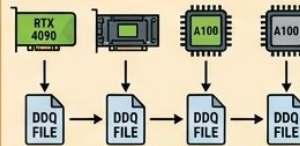
⚠ POOR CALIBRATION RISKS

- Slower Performance (vs FP16)
- Bad QDQ Node Placement



DEPLOYMENT REQUIREMENTS

GPU ARCHITECTURE SPECIFIC ENGINES



NO CROSS-HARDWARE PORTABILITY

⚠ MUST BUILD SEPARATE ENGINES
FOR EACH TARGET GPU

✗ POOR ACCURACY
(15-20% DEGRADATION)

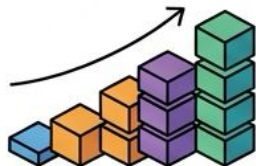
⚠ POOR CALIBRATION RISKS

- Slower Performance (vs FP16)
- Bad QDQ Node Placement



KV Cache

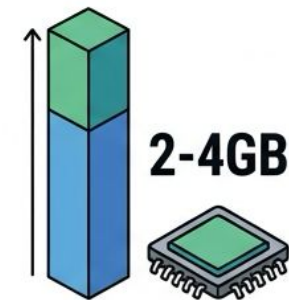
CONTINUOUS CACHE GROWTH



EACH NEW TOKEN
ATTENDS TO ALL
PREVIOUS TOKENS

CACHE GROWS WITH
EVERY OUTPUT TOKEN

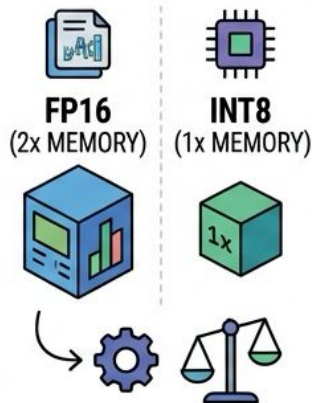
MEMORY USAGE (7B MODEL AT FP16)



KV CACHE
PER REQUEST

7B MODEL AT FP16
USES SIGNIFICANT
MEMORY PER REQUEST

INT8 KV CACHE QUANTIZATION



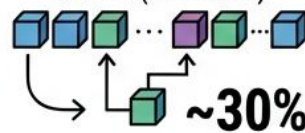
CUTS MEMORY **2x**
WITH NEGLIGIBLE
QUALITY IMPACT

PAGED ATTENTION ALLOCATION

CONTIGUOUS (WASTEFUL)

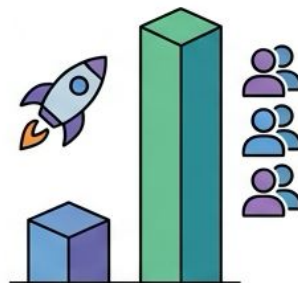


PAGED (EFFICIENT)



IMPROVES MEMORY
UTILIZATION BY **~30%**

COMBINED OPTIMIZATIONS EFFECT



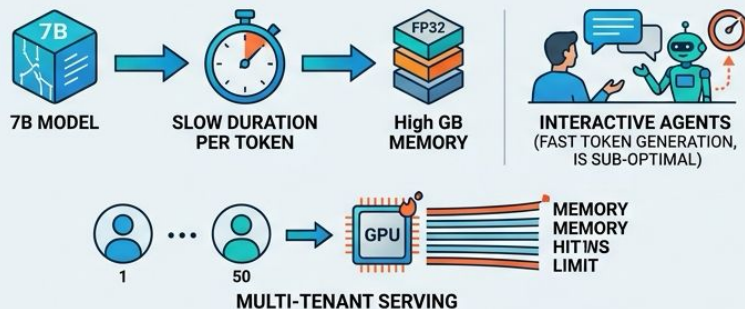
FP16
BASELINE

INT8
+ PAGED
ATTENTION

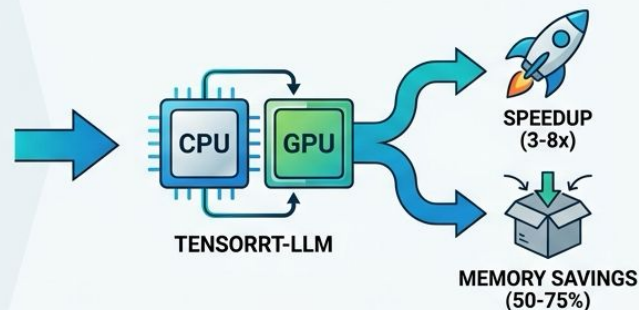
ENABLES **4-6x** MORE
CONCURRENT REQUESTS
ON SAME HARDWARE

Tensor Parallelism & TensorRT-LLM

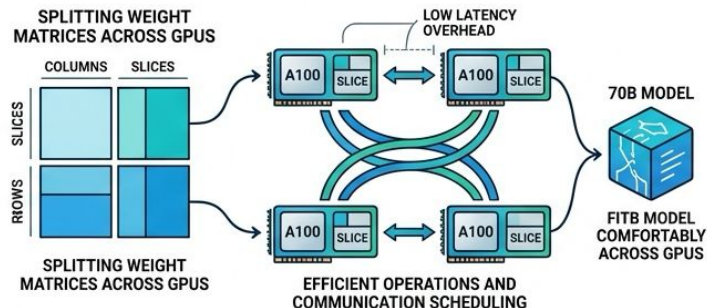
CHALLENGES IN PRODUCTION INFERENCE



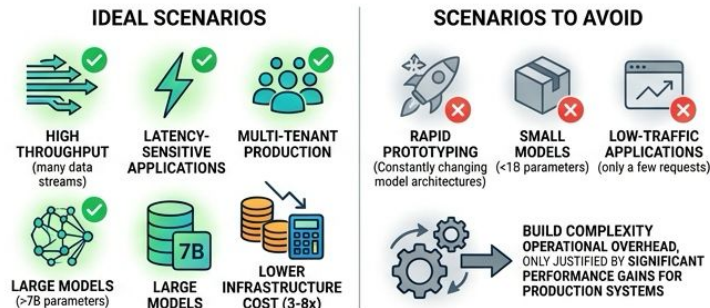
TENSORRT-LLM OPTIMIZATION



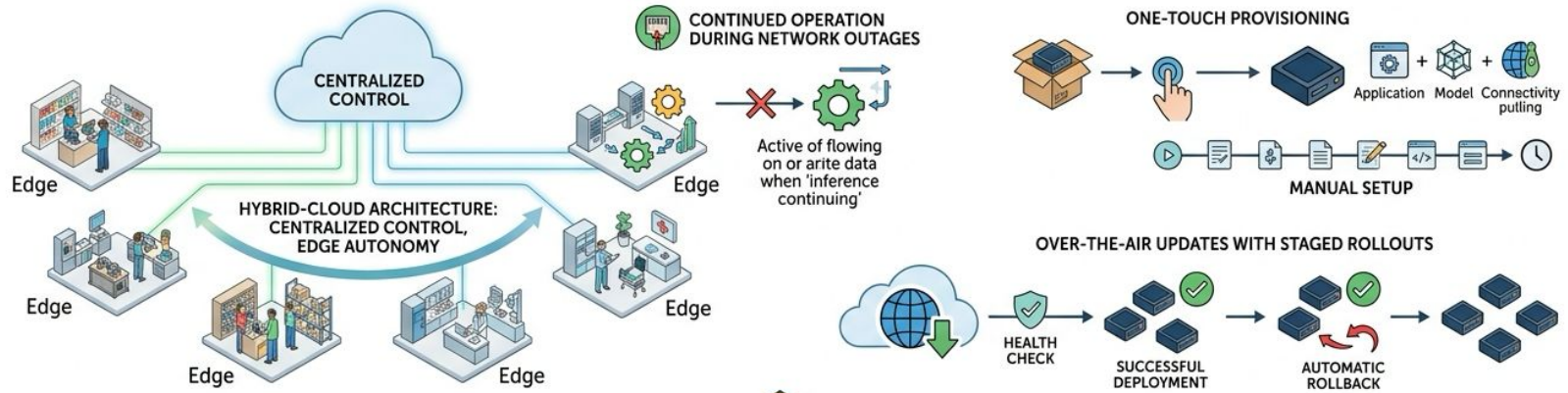
TENSOR PARALLELISM



WHEN TO APPLY TENSORRT-LLM



Fleet Command



Fleet Command Security -- Zero Trust at the Edge



SECURE BOOT



CERTIFICATE-BASED AUTHENTICATION WITH 24-48 HOUR ROTATION

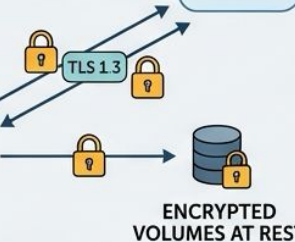


PRIVATE REGISTRY



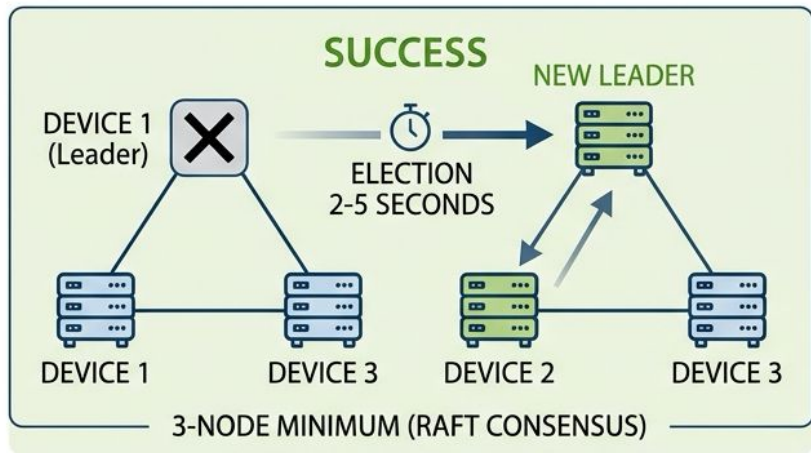
Any connect with authenticated edge nodes only

END-TO-END ENCRYPTION



variable physical security, with the on-device security keeping data safe

HIGH AVAILABILITY (HA)

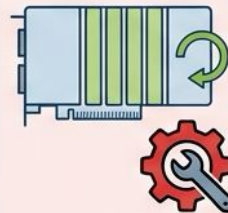
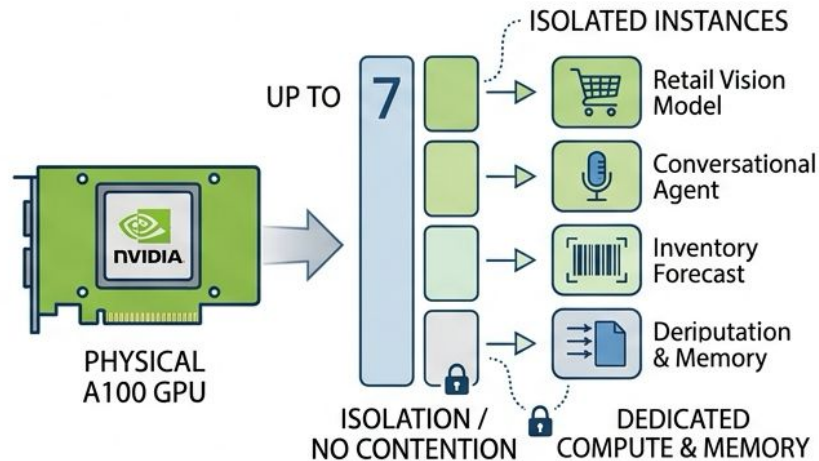


COMMON MISTAKE

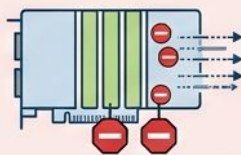


RAFT CONSENSUS: REQUIRES MAJORITY

MULTI-INSTANCE GPU (MIG)



MIG RECONFIGURATION:
DISRUPTIVE CHANGE



GPU RESET
KILLS ALL
WORKLOADS



SCHEDULE DURING
MAINTENANCE
WINDOWS ONLY

Worked Example -- 500-Store Retail Edge Deployment

- Pilot 25 stores first, then roll out to remaining 475
 - Staged deployments: 20% first, validate, then proceed
 - Automated health checks trigger rollback on failure
 - Weekly model updates: 50 locations per hour, fully automated
 - Configuration drift drops from 15-20% to under 1%
- 

When to Use What -- Decision Framework

TENSORRT-LLM

✓ WHEN TO USE



Production
Serving



Cost
Pressure



Latency
Pressure

✓ KEY SCENARIOS



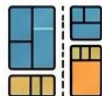
Cost Dominates
Budget



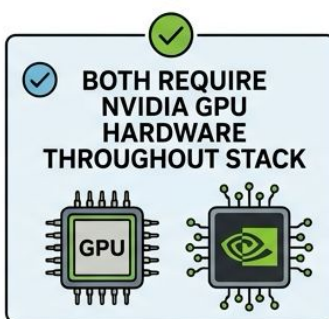
Low Token
Latency



Multi-Tenant
Deployment



Memory
Efficiency



OVERALL INSIGHT

FLEET COMMAND

✓ WHEN TO USE



50+ Edge Locations



Centralized
Management

✓ KEY SCENARIOS



Geographically
Distributed



No Local
Tech Staff



Over-the-Air
Updates



High Availability
& Audit Trails

✗ WHEN NOT TO USE



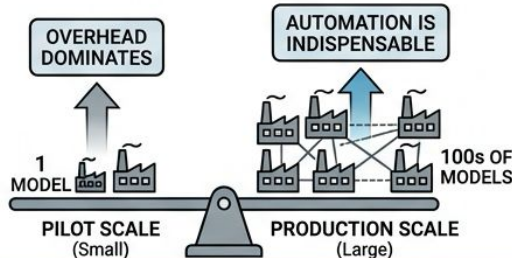
Prototyping



Frequent
Model Swaps



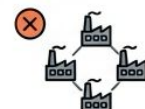
Small Model with
Good Latency



✗ WHEN NOT TO USE



Cloud-only
Workloads



Few Locations
(< 10)



Static
Models



TRADING UPFRONT COMPLEXITY FOR OPERATIONAL EFFICIENCY AT SCALE

Key Takeaways and Bridge to Chapter 4.7

- TensorRT-LLM stacks precision, fusion, KV cache, and batching for 3-8x gains
 - Calibration dataset quality is the single biggest INT8 success factor
 - Fleet Command turns edge sprawl into a managed, secure platform
 - Three-node minimum for HA; MIG changes only in maintenance windows
 - Next chapter: horizontal scaling strategies for growing user demand
- 